

DIT 204: Programming with Python

Introduction to Python Programming

What is Python?

Python is a high-level, interpreted, and general-purpose programming language used for developing software applications, automating tasks, analyzing data, and solving computational problems.

Python was created by Guido van Rossum and released in 1991.

Python is one of the most popular programming languages in the world because it is:

- Simple
- Easy to learn
- Powerful
- Flexible

Why Learn Python?

Python is widely used in:

- Web development
- Artificial Intelligence (AI)
- Data Science
- Machine Learning
- Cybersecurity
- Networking
- Automation
- Mobile applications
- Scientific computing

1.3 Features of Python

1. Simple and Easy to Learn

Python syntax is easy to read and understand.

Example:

```
print("Hello World")
```

2. Interpreted Language

Python executes programs line by line using an interpreter.

3. High-Level Language

Python is close to human language and easier than low-level languages.

4. Portable

Python programs can run on:

- Windows
- Linux
- MacOS

without changing the code.

5. Open Source

Python is free to use and distribute.

6. Object-Oriented

Python supports object-oriented programming concepts such as:

- Classes
- Objects
- Inheritance

7. Large Standard Library

Python has many built-in modules and libraries.

Examples:

- math
- random
- datetime

Applications of Python

Area	Use
Web Development	Building websites
Artificial Intelligence	Smart systems
Data Science	Data analysis
Cybersecurity	Security testing
Automation	Task automation
Networking	Network scripting
Education	Teaching programming
Game Development	Building games

Advantages of Python

1. Easy syntax
2. Faster development
3. Large community support
4. Cross-platform compatibility
5. Supports modern technologies like AI and Machine Learning

Limitations of Python

1. Slower execution compared to compiled languages
2. High memory consumption
3. Not suitable for some mobile applications

Python Syntax

Python syntax refers to the rules used to write Python programs.

Example:

```
print("Welcome to Python")
```

Output:

Welcome to Python

1.8 Rules of Python Syntax

1. Indentation

Python uses indentation to define blocks of code.

Example:

```
if 5 > 2:
    print("Five is greater")
```

2. Case Sensitivity

Python is case-sensitive.

Example:

```
name = "John"
Name = "Peter"
```

name and Name are different variables.

3. Comments

Comments are used to explain code.

Single-line Comment

```
# This is a comment
```

Writing the First Python Program

Example 1: Hello World Program

```
print("Hello World")
```

Explanation

- `print()` displays output on the screen.

Example 2: Displaying a Name

```
print("My name is Ibrahim")
```

Basic Programming Terminologies

Term	Meaning
Program	A set of instructions
Bug	An error in a program
Debugging	Finding and fixing errors
Syntax	Rules of writing code
Variable	A storage location
Interpreter	Executes Python code
Compiler	Converts code into machine language

Python Interpreter

A Python interpreter is software that reads and executes Python code line by line.

Examples:

- Python IDLE
- VS Code
- PyCharm
- Jupyter Notebook

Difference Between Compiler and Interpreter

Compiler

Translates whole program at once

Faster execution

Errors shown after compilation

Python uses an interpreter.

Interpreter

Translates line by line

Slower execution

Errors shown immediately

Sample Python Programs

Program 1: Addition

```
a = 10
b = 5

sum = a + b

print(sum)
```

Output

15

Program 2: Multiplication

```
x = 4
y = 3

result = x * y

print(result)
```

Output

12

Areas Where Python is Used in Real Life

Banking

- ATM systems
- Fraud detection

Education

- Student management systems

Social Media

- Instagram
- YouTube
- TikTok

Health

- Hospital management systems

E-commerce

- Online shopping platforms

Summary

In this lesson, we look at:

- Meaning of Python programming
- Features of Python
- Applications of Python
- Advantages and limitations of Python
- Python syntax
- Basic Python programs
- Important programming terminologies

Class Activities

Activity 1

Write a Python program to display your:

- Name
- Program
- Level

Activity 2

Identify whether the following are syntax errors or not:

`print("Hello)`

`Print("Welcome")`

Trial questions

1. Define Python programming.
2. State five features of Python.
3. Mention five applications of Python.
4. Explain the difference between a compiler and an interpreter.
5. Write a Python program to display:

Welcome to DIT 204

Suggested Reading

1. Python Crash Course – Eric Matthes
2. Automate the Boring Stuff with Python – Al Sweigart
3. Introduction to Python Programming – John Zelle